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The Editors
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Elsevier

Dear Editors,

We submit for your consideration the manuscript entitled **"WaveSplat: Grid-Free Wave Field Estimation from Sparse 3D Point Clouds with a Learned Spectral Prior"** for publication in *Ocean Engineering*.

Scope and motivation. Real-time wave field estimation from sparse 3-D point clouds is a critical enabler for autonomous maritime operations, including proactive motion compensation, route optimisation, and wave-preview feedforward control. Existing methods rely on spatial grid discretisation combined with ensemble Kalman filtering, which introduces interpolation artefacts and fixed spatial resolution. This submission addresses these limitations with a fundamentally different representation.

Principal contributions. WaveSplat represents the ocean wave field as a superposition of spectral components — plane waves constrained by the linear dispersion relation — optimised directly on sparse, irregularly distributed LiDAR observations without any spatial grid. Four novel contributions are presented:

1. A *grid-free spectral estimation framework* with persistent cross-frame tracking via dispersion-based phase prediction, enabling real-time operation with approximately 10 gradient steps per frame.
2. A *dispersion-constrained parameterisation* that reduces the free parameter count from $5N$ to $4N$. **[SCOPE 2026-08-14]** The accuracy contribution of this constraint has *not yet been measured*; the ablation previously cited removed the frequency initialisation along with the constraint.
3. **[RETRACTED 2026-08-14]** An *incidence-angle-weighted observation model* was presented as a contribution; on a fresh out-of-sample corpus, disabling it was *better* in both strata, so it is withdrawn as a contribution and retained only as a negative result.
4. **[RETRACTED 2026-08-14 as a contribution]** An *adaptive model complexity* mechanism selects the number of spectral components from N=1 for simple swells to N=21 for complex multi-component seas. Its *accuracy* benefit was subsequently measured as null; it is reported as a mechanism only.

Key results. On a physics-based simulation dataset of 100 episodes spanning Beaufort 2–5 sea states (1–30 wave components), WaveSplat achieves: **[RETRACTED 2026-08-14]** The accuracy, latency, Skill Score and significance figures previously listed here are **withdrawn in full**. In the underlying result file the proposed method and the baselines were never scored on the same frames; the verdict reverses with the choice of baseline (against a tuned LETKF the same column is a statistical null); the latency figures were measured in separate processes; and the corpus does not contain enough independent draws to support the quoted p-values. No replacement numbers are stated, because none have been verified for this comparison. The only

citable latency statement is 1.5–2.6× within one process on a single thread (CPU: AMD Ryzen 7 7800X3D).

Relevance to Ocean Engineering. This work directly addresses the operational challenge of real-time wave field reconstruction for vessel-scale maritime applications. The method requires no pre-trained models, no offline calibration beyond water depth, and runs on standard CPU hardware at 10-Hz LiDAR frame rates. Results include a companion application to wave-preview feedforward motion control of hydrofoil vessels, demonstrating the practical value of the spectral representation for downstream maritime control systems.

Originality statement. This work is original and is not under consideration elsewhere. The simulation code, all experiment scripts, and five representative episodes are openly available at the project repository to ensure end-to-end reproducibility of all reported tables and figures.

Suggested reviewers. We suggest the following experts in ocean wave sensing and data assimilation (the authors have no conflict of interest with any of them):

- Prof. Tianhao Liu (wave sensing from ship-mounted sensors)
- Prof. Johannes Gemmrich (ocean surface wave observations)
- Dr. Nicolas Desmars (LiDAR-based wave field reconstruction)

We look forward to your consideration of this manuscript.

Sincerely,

Saeyong Park